MetroMart Retail Group — Client Data Project

Confidential simulated client material

Background

MetroMart Retail Group is a fictional South African general-retail business operating stores across Gauteng. The company has accumulated operational data from several exports and systems. Management currently struggles to produce consistent reporting because the source data is fragmented and contains quality issues.

Client Request

Review the supplied data, identify and document data-quality issues, clean and validate the data, analyse the available information, and design a relational database that can support reliable operational and analytical reporting.

Expected Outcome

The client expects a documented transformation from raw source data into a structured database supporting customer, product, store, sales and returns information and allowing management questions to be answered with SQL.

Important

Treat the supplied files as if they were received from a real client. Do not assume the source data is correct. Record assumptions, decisions, unresolved questions and validation checks. The client has not supplied a database schema.

Initial Business Questions

Management would eventually like to answer questions such as:

• Which stores and product categories generate the most revenue?

• How have sales changed over time?

• Which products perform strongly or poorly?

• Which customers purchase most frequently?

• How significant are returns?

• Are there data-quality problems that could affect reporting?